

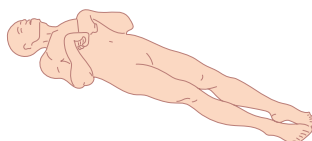
► NEUROLOGY—PATHOLOGY

Common brain lesions

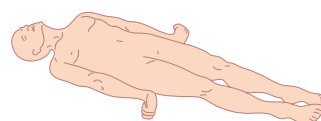
AREA OF LESION	COMPLICATIONS
Prefrontal cortex	Frontal lobe syndrome —disinhibition, hyperphagia, impulsivity, loss of empathy, impaired executive function, akinetic mutism. Seen in frontotemporal dementia.
Frontal eye fields	Eyes look toward brain lesion (ie, away from side of hemiplegia). Seen in MCA stroke.
Paramedian pontine reticular formation	Eyes look away from brain lesion (ie, toward side of hemiplegia).
Dominant parietal cortex	Gerstmann syndrome —agraphia, acalculia, finger agnosia, left-right disorientation.
Nondominant parietal cortex	Hemispatial neglect syndrome —agnosia of the contralateral side of the world.
Basal ganglia	Tremor at rest, chorea, athetosis. Seen in Parkinson disease, Huntington disease.
Subthalamic nucleus	Contralateral hemiballismus.
Mammillary bodies	Bilateral lesions → Wernicke-Korsakoff syndrome (due to thiamine deficiency).
Amygdala	Bilateral lesions → Klüver-Bucy syndrome —disinhibition (eg, hyperphagia, hypersexuality, hyperorality). Seen in HSV-1 encephalitis.
Hippocampus	Bilateral lesions → anterograde amnesia (no new memory formation). Seen in Alzheimer disease.
Dorsal midbrain	Parinaud syndrome (often due to pineal gland tumors).
Reticular activating system	Reduced levels of arousal and wakefulness, coma.
Medial longitudinal fasciculus	Internuclear ophthalmoplegia (impaired adduction of ipsilateral eye; nystagmus of contralateral eye with abduction). Seen in multiple sclerosis.
Cerebellar hemisphere	Intention tremor, limb ataxia, loss of balance; damage to cerebellum → ipsilateral deficits; fall toward side of lesion. Cerebellar hemispheres are laterally located—affect lateral limbs.
Cerebellar vermis	Truncal ataxia (wide-based, “drunken sailor” gait), nystagmus, dysarthria. Degeneration associated with chronic alcohol overuse. Vermis is centrally located—affects central body.

Abnormal motor posturing

	Decorticate (flexor) posturing	Decerebrate (extensor) posturing
SITE OF LESION	Above red nucleus (often cerebral cortex)	Between red and vestibular nuclei (brainstem)
OVERACTIVE TRACTS	Rubrospinal and vestibulospinal tracts	Vestibulospinal tract
PRESENTATION	Upper limb flexion, lower limb extension	Upper and lower limb extension
NOTES	“Your hands are near the cor (heart)”	Worse prognosis



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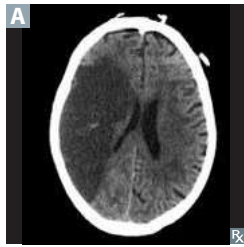
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Ischemic brain disease/stroke

Irreversible neuronal injury begins after 5 minutes of hypoxia. Most **vulnerable: hippocampus** (CA1 region), **neocortex**, cerebellum (**Purkinje cells**), **watershed areas** (“**vulnerable hippos need pure water**”).

Stroke imaging: noncontrast CT to exclude hemorrhage (before tPA can be given). CT detects ischemic changes in 6–24 hr. Diffusion-weighted MRI can detect ischemia within 3–30 min.

TIME SINCE ISCHEMIC EVENT	12–24 HOURS	24–72 HOURS	3–5 DAYS	1–2 WEEKS	> 2 WEEKS
Histologic features	Eosinophilic cytoplasm + pyknotic nuclei (red neurons)	Necrosis + neutrophils	Macrophages (microglia)	Reactive gliosis (astrocytes) + vascular proliferation	Glial scar

Ischemic stroke

Ischemia → infarction → liquefactive necrosis.

3 types:

- **Thrombotic**—due to a clot forming directly at site of infarction (commonly the MCA **A**), usually over a ruptured atherosclerotic plaque.
- **Embolic**—due to an embolus from another part of the body. Can affect multiple vascular territories. Examples: atrial fibrillation, carotid artery stenosis, DVT with patent foramen ovale (paradoxical embolism), infective endocarditis.
- **Hypoxic**—due to systemic hypoperfusion or hypoxemia. Common during cardiovascular surgeries, tends to affect watershed areas.

Treatment: tPA (if within 3–4.5 hr of onset and no hemorrhage/risk of hemorrhage) and/or thrombectomy (if large artery occlusion). Reduce risk with medical therapy (eg, aspirin, clopidogrel); optimum control of blood pressure, blood sugars, lipids; smoking cessation; and treat conditions that ↑ risk (eg, atrial fibrillation, carotid artery stenosis).

Transient ischemic attack

Brief, reversible episode of focal neurologic dysfunction without acute infarction (⊖ MRI), with the majority resolving in < 15 minutes; ischemia (eg, embolus, small vessel stenosis). May present with amaurosis fugax (transient visual loss) due to retinal artery emboli from carotid artery disease.

Cerebral edema

Fluid accumulation in brain parenchyma → ↑ ICP. Types:

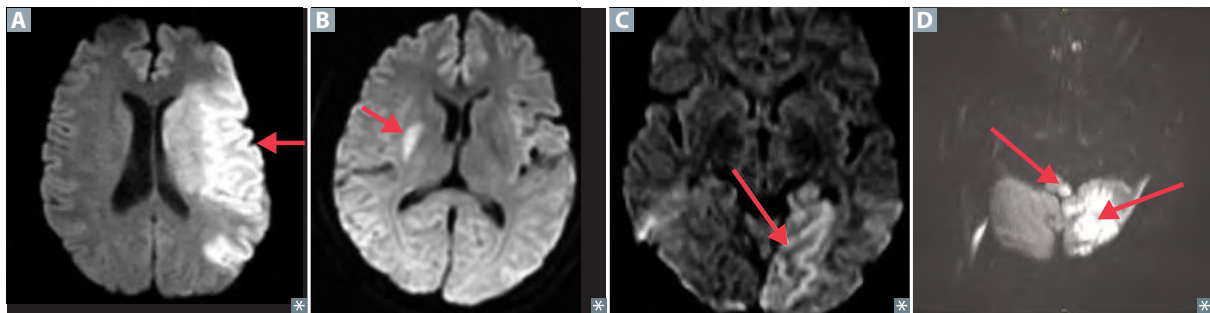
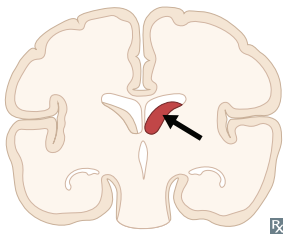
- **Cytotoxic edema**—intracellular fluid accumulation due to osmotic shift (eg, Na^+/K^+ -ATPase dysfunction → ↑ intracellular Na^+). Caused by ischemia (early), hyperammonemia, SIADH.
- **Vasogenic edema**—extracellular fluid accumulation due to disruption of BBB (↑ permeability). Caused by ischemia (late), trauma, hemorrhage, inflammation, tumors (arrows in **A** show surrounding vasogenic edema).

Effects of strokes

ARTERY	AREA OF LESION	SYMPTOMS	NOTES
Anterior circulation			
Anterior cerebral artery	Motor and sensory cortices—lower limb.	Contralateral paralysis and sensory loss—lower limb, urinary incontinence.	
Middle cerebral artery	Motor and sensory cortices A —upper limb and face. Temporal lobe (Wernicke area); frontal lobe (Broca area).	Contralateral paralysis and sensory loss—lower face and upper limb. Aphasia if in dominant (usually left) hemisphere. Hemineglect if lesion affects nondominant (usually right) hemisphere.	Wernicke aphasia is associated with right superior quadrant visual field defect due to temporal lobe involvement.
Lenticulo-striate artery	Striatum, internal capsule.	Contralateral paralysis. Absence of cortical signs (eg, neglect, aphasia, visual field loss).	Pure motor stroke (most common). Common location of lacunar infarcts B , due to microatheroma and hyaline arteriosclerosis (lipohyalinosis) 2° to unmanaged hypertension.
Posterior circulation			
Posterior cerebral artery	Occipital lobe C .	Contralateral hemianopia with macular sparing; alexia without agraphia (dominant hemisphere, extending to splenium of corpus callosum); prosopagnosia (inability to recognize familiar faces; nondominant hemisphere).	Weber syndrome —midbrain stroke due to occlusion of paramedian branches of PCA → ipsilateral CN III palsy and contralateral hemiplegia (damage to ipsilateral cerebral peduncle).
Basilar artery	Pons, medulla, lower midbrain. Corticospinal and corticobulbar tracts. Ocular cranial nerve nuclei, paramedian pontine reticular formation.	If RAS spared, consciousness is preserved. Quadriplegia ; loss of voluntary facial (except blinking), mouth, and tongue movements. Loss of horizontal, but not vertical, eye movements.	Locked-in syndrome (locked in the basement).
Anterior inferior cerebellar artery	Facial nerve nuclei. Vestibular nuclei. Spinothalamic tract, spinal trigeminal nucleus. Sympathetic fibers. Middle and inferior cerebellar peduncles. Inner ear.	Paralysis of face (LMN lesion vs UMN lesion in cortical stroke), ↓ lacrimation, ↓ salivation, ↓ taste from anterior 2/3 of tongue. Vomiting, vertigo, nystagmus ↓ pain and temperature sensation from contralateral body, ipsilateral face. Ipsilateral Horner syndrome. Ipsilateral ataxia, dysmetria. Ipsilateral sensorineural deafness, vertigo.	Lateral pontine syndrome. Facial nerve nuclei effects are specific to AICA lesions. Supplied by labyrinthine artery, a branch of AICA.

Effects of strokes (continued)

ARTERY	AREA OF LESION	SYMPTOMS	NOTES
Posterior inferior cerebellar artery	Nucleus ambiguus (CN IX, X).	Dysphagia, hoarseness , ↓ gag reflex, hiccups.	Lateral medullary (Wallenberg) syndrome. Nucleus ambiguus effects are specific to PICA lesions D . “Don’t pick a (PICA) lame (lateral medullary syndrome) horse (hoarseness) that can’t eat (dysphagia).”
	Vestibular nuclei.	Vomiting, vertigo, nystagmus	
	Lateral spinothalamic tract, spinal trigeminal nucleus.	↓ pain and temperature sensation from contralateral body, ipsilateral face.	
	Sympathetic fibers. Inferior cerebellar peduncle.	Ipsilateral Horner syndrome. Ipsilateral ataxia, dysmetria.	
Anterior spinal artery	Corticospinal tract.	Contralateral paralysis—upper and lower limbs.	Medial Medullary syndrome —caused by infarct of paramedian branches of ASA and/or vertebral arteries. Ants love M&M’s .
	Medial lemniscus.	↓ contralateral proprioception.	
	Caudal medulla—hypoglossal nerve.	Ipsilateral hypoglossal dysfunction (tongue deviates ipsilaterally).	

**Neonatal intraventricular hemorrhage**

Bleeding into ventricles (arrow in illustration shows blood in intraventricular space). ↑ risk in premature and low-birth-weight infants. Originates in germinal matrix, a highly vascularized layer within the subventricular zone. Due to reduced glial fiber support and impaired autoregulation of BP in premature infants. Can present with altered level of consciousness, bulging fontanelle, hypotension, seizures, coma.

Extracranial injuries

Occur during birth leading to blood accumulation within the scalp and skull. Commonly seen in vacuum-assisted delivery.

DISORDER	PRESENTATION
Caput succedaneum	Self-limited, benign, edematous swelling above periosteum; crosses suture lines. May be caused by prolonged fetal–birth canal engagement. Resolves spontaneously.
Subgaleal hemorrhage	Serious, life-threatening damage of fetal emissary veins → blood accumulation between periosteum and galea aponeurosis. Presents as diffuse, fluctuant scalp swelling extending posteriorly and laterally. May lead to anemia and hypovolemic shock.
Cephalohematoma	Blood accumulation between skull and periosteum; does not cross suture lines. May be caused by forcep delivery. Presents with firm, localized swelling over parietal or occipital lobe. May lead to indirect hyperbilirubinemia.

Intracranial hemorrhage**Epidural hematoma**

Rupture of middle meningeal artery (branch of maxillary artery), often 2° to skull fracture (circle in **A**) involving the pterion (thinnest area of the lateral skull). Might present with transient loss of consciousness → recovery (“lucid interval”) → rapid deterioration due to hematoma expansion. Scalp hematoma (arrows in **A**) and rapid intracranial expansion (arrows in **B**) under systemic arterial pressure → transtentorial herniation, CN III palsy. CT shows biconvex (lenticiform), hyperdense blood collection **B** not crossing suture lines.

Subdural hematoma

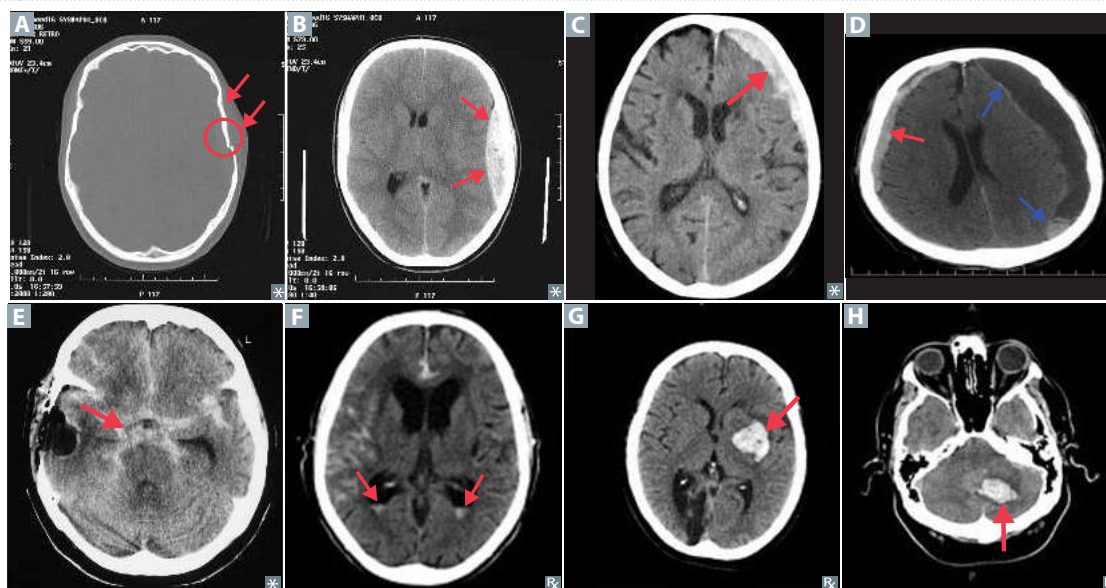
Rupture of bridging veins. Can be acute (traumatic, high-energy impact → brighter, or hyperdense, on CT) or chronic (associated with mild trauma, cerebral atrophy, ↑ age, chronic alcohol overuse → darker, or hypodense, on CT). Also seen in shaken babies. Crescent-shaped hemorrhage (red arrows in **C** and **D**) that **crosses suture lines**. Can cause midline shift, findings of acute (hyperdense border) on chronic (hypodense crescent) hemorrhage (blue arrows in **D**).

Subarachnoid hemorrhage

Bleeding **E F** due to trauma, or rupture of an aneurysm (such as a saccular aneurysm) or arteriovenous malformation. Rapid time course. Patients complain of “worst headache of my life.” Bloody or yellow (xanthochromic) LP. Vasospasm can occur due to blood breakdown or rebleed 3–10 days after hemorrhage → ischemic infarct; nimodipine used to prevent/reduce vasospasm. ↑ risk of developing communicating and/or noncommunicating hydrocephalus.

Intraparenchymal hemorrhage

Most commonly caused by systemic hypertension. Also seen with amyloid angiopathy (recurrent lobar hemorrhagic stroke in older adults), arteriovenous malformations, vasculitis, neoplasm. May be 2° to reperfusion injury in ischemic stroke. Hypertensive hemorrhages (Charcot-Bouchard microaneurysm) most often occur in putamen/globus pallidus of basal ganglia (lenticulostriate vessels **G**), followed by internal capsule, thalamus, pons, and cerebellum **H**.



Thalamic pain syndrome Severe, treatment-resistant neuropathic pain following thalamic lesions; may be due to occlusion of a lenticulostriate artery. Initial paresthesias followed in weeks to months by allodynia (ordinarily painless stimuli cause pain), hyperalgesia (hypersensitivity to pain), and dysesthesia (altered sensation) on the contralateral side.

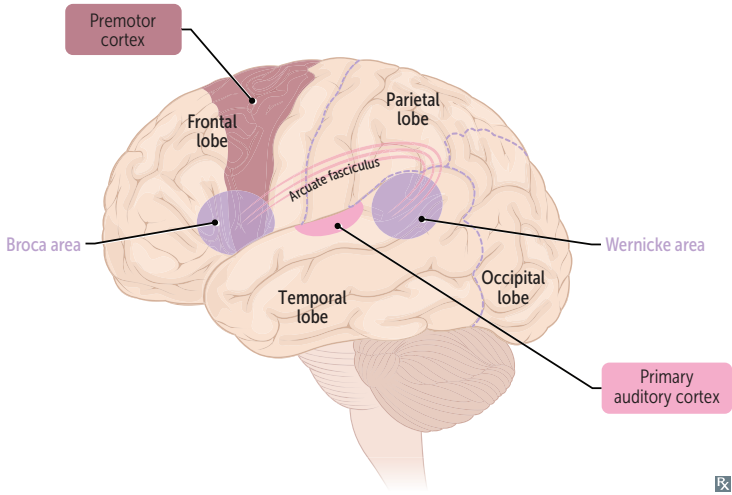
Phantom limb pain Sensation of burning, aching, or electric shock–like pain in a limb that is no longer present. Common after amputation. Associated with reorganization of the 1° somatosensory cortex.

Diffuse axonal injury Traumatic shearing of white matter tracts during rapid acceleration and/or deceleration of the brain (eg, motor vehicle accident). Usually results in devastating neurologic injury, often causing coma or persistent vegetative state. MRI shows multiple lesions (punctate hemorrhages) involving white matter tracts **A**.



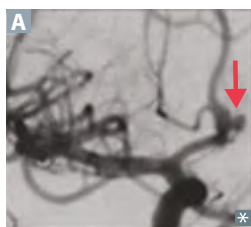
Aphasia Higher-order language deficit (inability to understand/produce/use language appropriately); caused by pathology in dominant cerebral hemisphere (usually left). Distinguish from **dysarthria**—motor inability to produce speech (movement deficit).

TYPE	COMMENTS
Broca (expressive)	Broca area in inferior frontal gyrus of frontal lobe. Nonfluent speech with intact language comprehension. Patients appear frustrated, insight intact. Broca = broken boca (<i>boca</i> = mouth in Spanish).
Wernicke (receptive)	Wernicke area in superior temporal gyrus of temporal lobe. Fluent speech with impaired language comprehension. Patients do not have insight. Wernicke is a word salad and makes no sense.
Conduction	Can be caused by damage to arcuate fasciculus. Impaired speech repetition.
Global	Broca and Wernicke areas affected. Nonfluent speech with impaired language comprehension.



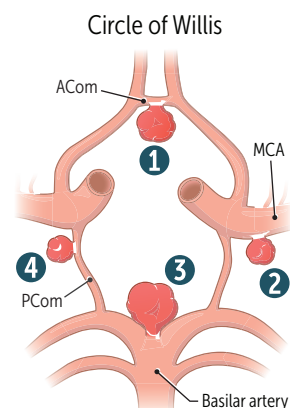
Aneurysms

Abnormal dilation of an artery due to weakening of vessel wall.

Saccular aneurysm

Also called berry aneurysm **A**. Occurs at bifurcations in the circle of Willis. Most common site is junction of ACom and ACA. Associated with ADPKD, Ehlers-Danlos syndrome. Other risk factors: advanced age, hypertension, tobacco smoking. Usually clinically silent until rupture (most common complication) → subarachnoid hemorrhage (“worst headache of my life” or “thunderclap headache”) → focal neurologic deficits. Can also cause symptoms via direct compression of surrounding structures by growing aneurysm.

- ACom—compression → bitemporal hemianopia (compression of optic chiasm); visual acuity deficits; rupture → ischemia in ACA distribution → contralateral lower extremity hemiparesis, sensory deficits.
- MCA—rupture → ischemia in MCA distribution → contralateral upper extremity and lower facial hemiparesis, sensory deficits.
- PCom—compression → ipsilateral CN III palsy → mydriasis (“blown pupil”); may also see ptosis, “down and out” eye.



- 1 Anterior communicating artery (ACom) aneurysm
- 2 Middle cerebral artery (MCA) aneurysm
- 3 Basilar tip aneurysm
- 4 Posterior communicating artery (PCom) aneurysm

Charcot-Bouchard microaneurysm

Common, associated with chronic hypertension; affects small vessels (eg, lenticulostriate arteries in basal ganglia, thalamus) and can cause hemorrhagic intraparenchymal strokes. Not visible on angiography.

Fever vs heat stroke

	Fever	Heat stroke
PATHOPHYSIOLOGY	Cytokine activation during inflammation (eg, infection)	Inability of body to dissipate heat (eg, exertion)
TEMPERATURE	Usually < 40°C (104°F)	Usually > 40°C (104°F)
COMPLICATIONS	Febrile seizure (benign, usually self-limiting)	CNS dysfunction (eg, confusion), rhabdomyolysis, acute kidney injury, ARDS, DIC
MANAGEMENT	Acetaminophen or ibuprofen for comfort (does not prevent future febrile seizures), antibiotic therapy if indicated	Rapid external cooling, rehydration and electrolyte correction

Seizures

Characterized by synchronized, high-frequency neuronal firing. Consist of 3 phases:

- **Aura**—early part of a seizure, may include odd smells or tastes.
- **Ictal**—time from first symptom to end of seizure activity.
- **Postictal**—period of gradual recovery back to preseizure baseline level of function/awareness.

Focal seizures

Originate in a single area of the brain, most commonly the medial temporal lobe. Types:

- **Focal aware** (formerly called simple partial)—consciousness intact; motor, sensory, autonomic, psychic symptoms
- **Focal impaired awareness** (formerly called complex partial)—impaired consciousness, automatisms

Generalized seizures

Diffuse. Types:

- **Absence** (petit mal)—3 Hz spike-and-wave discharges on EEG; short (usually 10 seconds), frequent episodes of blank stare, possible automatisms; no postictal confusion. Can be triggered by hyperventilation
- **Myoclonic**—quick, repetitive jerks; no loss of consciousness or postictal confusion
- **Tonic-clonic** (grand mal)—alternating stiffening and movement, postictal confusion, urinary incontinence, tongue biting
- **Tonic**—stiffening
- **Atonic**—“drop” seizures (falls to floor); commonly mistaken for fainting

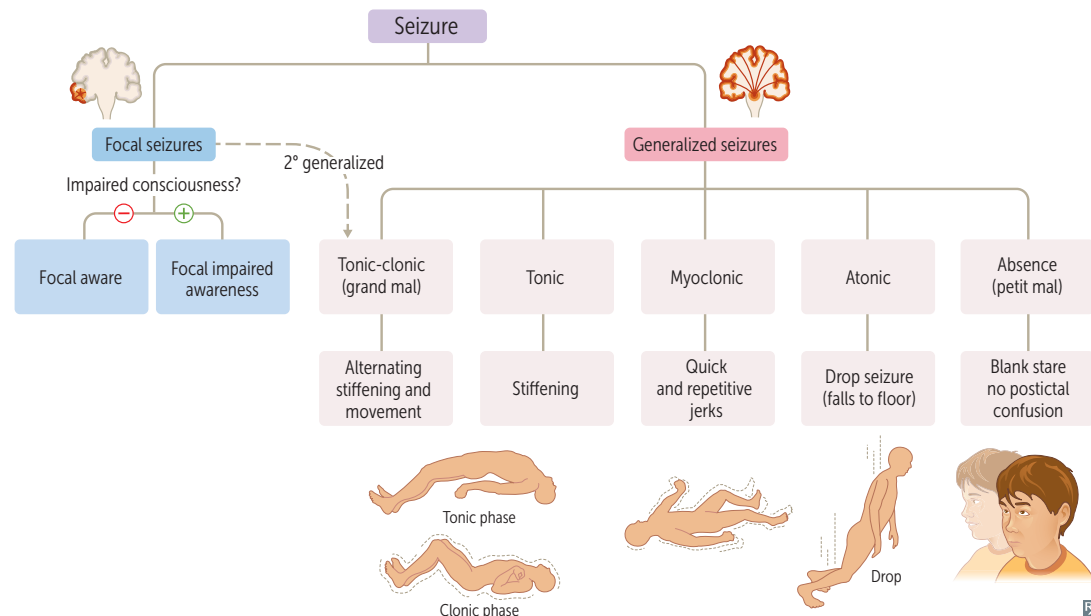
Epilepsy—disorder of recurrent, unprovoked seizures (febrile seizures are not epilepsy).

Convulsive status epilepticus—continuous (≥ 5 min) or recurring seizures without interictal return to baseline consciousness that may result in brain injury.

Causes of seizures by age:

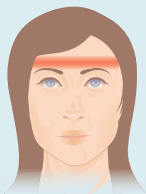
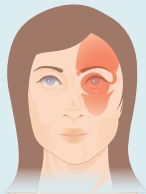
- Children < 18 —genetic, infection (febrile), trauma, congenital, metabolic
- Adults 18–65—tumor, trauma, stroke, infection
- Adults > 65 —stroke, tumor, trauma, metabolic, infection

Psychogenic nonepileptic events—resemble prolonged (> 1 minute) syncopal or tonic-clonic episodes without postictal phase, autonomic disturbances, or tongue biting. Often witnessed with vocalizations and preceding aura. Female sex predominance. Risk factors: history of psychiatric disorders, substance use. Normal video EEG.



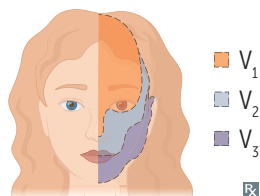
Headaches

Pain due to irritation of intra- or extracranial structures (eg, meninges, blood vessels). Primary headaches include tension-type, migraine, and cluster. Secondary headaches include medication overuse, meningitis, subarachnoid hemorrhage, hydrocephalus, neoplasia, giant cell arteritis.

CLASSIFICATION	LOCALIZATION	DURATION	DESCRIPTION	TREATMENT
Tension-type 	Bilateral	> 30 min (typically 4–6 hr); constant	Steady, “bandlike” pain. No nausea or vomiting. No more than one of photophobia or phonophobia. No aura. Most common primary headache; more common in females.	Acute: analgesics, NSAIDs, acetaminophen. Prophylaxis: TCAs (eg, amitriptyline), behavioral therapy.
Migraine 	Unilateral	4–72 hr	Pulsating pain with nausea, photophobia, and/or phonophobia. May have “aura.” Due to irritation of CN V, meninges, or blood vessels (release of vasoactive neuropeptides [eg, substance P, calcitonin gene-related peptide]). More common in females. POUND —Pulsatile, One-day duration, Unilateral, Nausea, Disabling.	Acute: NSAIDs, triptans, dihydroergotamine, antiemetics (eg, prochlorperazine, metoclopramide). Prophylaxis: lifestyle changes (eg, sleep, exercise, diet), β -blockers, amitriptyline, topiramate, valproate, botulinum toxin, anti-CGRP monoclonal antibodies.
Cluster 	Unilateral	15 min–3 hr; repetitive	Excruciating periorbital pain with autonomic symptoms (eg, lacrimation, rhinorrhea, conjunctival injection). May present with Horner syndrome. More common in males.	Acute: sumatriptan, 100% O ₂ . Prophylaxis: verapamil.

Trigeminal neuralgia

Recurrent brief episodes of intense unilateral pain in CN V distribution (usually V₂ and/or V₃). Most cases are due to compression of CN V root by an aberrant vascular loop. Pain is described as electric shock–like or stabbing and usually lasts seconds. Typically triggered by light facial touch or facial movements (eg, chewing, talking). Treatment: carbamazepine, oxcarbazepine.



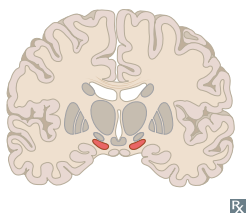
Dyskinesias

DISORDER	PRESENTATION	NOTES
Akathisia	Restlessness and intense urge to move	Can be seen with neuroleptic use or as an adverse effect of Parkinson disease treatment
Asterixis	“Flapping” motion upon extension of wrists	Associated with hepatic encephalopathy, Wilson disease, and other metabolic derangements
Athetosis	Slow, snakelike, writhing movements; especially seen in the fingers	Caused by lesion to basal ganglia Seen in Huntington disease
Chorea	Sudden, jerky, purposeless movements	<i>Chorea</i> (Greek) = dancing Caused by lesion to basal ganglia Seen in Huntington disease and acute rheumatic fever (Sydenham chorea).
Dystonia	Sustained, involuntary muscle contractions	Writers cramp, blepharospasm, torticollis Treatment: botulinum toxin injection
Essential tremor	High-frequency tremor with sustained posture (eg, outstretched arms); worsened with movement or anxiety	Often familial Patients often self-medicate with alcohol, which ↓ tremor amplitude Treatment: nonselective β -blockers (eg, propranolol), barbiturates (primidone)
Intention tremor	Slow, zigzag motion when pointing/extending toward a target	Caused by cerebellar dysfunction
Resting tremor	Uncontrolled movement of distal appendages (most noticeable in hands); tremor alleviated by intentional movement	Caused by lesion to substantia nigra Occurs at rest ; “pill-rolling tremor” of Parkinson disease; when you park your car, it is at rest
Hemiballismus	Sudden, wild flailing of one side of the body	Caused by lesion to contralateral subthalamic nucleus (eg, due to lacunar stroke) In hemiballismus , half -of-body is going ballistic
Myoclonus	Sudden, brief, uncontrolled muscle contraction	Jerks; hiccups; common in metabolic abnormalities (eg, renal and liver failure), Creutzfeldt-Jakob disease

Restless legs syndrome Uncomfortable sensations in legs causing irresistible urge to move them. Emerge during periods of inactivity; most prominent in the evening or at night. Transiently relieved by movement (eg, walking). Usually idiopathic (often with genetic predisposition), but may be associated with iron deficiency, CKD, diabetes mellitus (especially with neuropathy). Treatment: gabapentinoids, dopamine agonists.

Neurodegenerative movement disorders

Parkinson disease



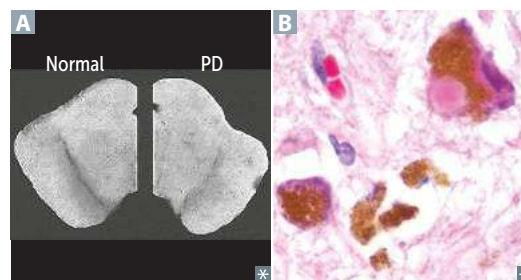
Loss of dopaminergic neurons in substantia nigra pars compacta (depigmentation in **A**). Symptoms typically manifest after age 60 (“body **TRAP**”):

- **T**remor (pill-rolling tremor at rest)
- **R**igidity (cogwheel or leadpipe)
- **A**kinesia/bradykinesia → shuffling gait, small handwriting (micrographia)
- **P**ostural instability (tendency to fall)

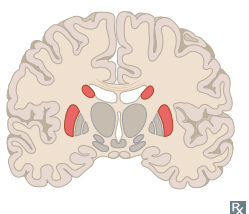
Dementia is usually a late finding.

Affected neurons contain Lewy bodies: intracellular eosinophilic inclusions composed of α -synuclein **B**. Think “Parkins**yn**uclein.”

Progressive supranuclear palsy—a Parkinson-plus syndrome. Clinical presentation: TRAP features, vertical gaze palsy, and cognitive dysfunction. Associated with “hummingbird sign” on midbrain MRI.



Huntington disease



Loss of GABAergic neurons in striatum. Autosomal dominant trinucleotide (CAG)_n repeat expansion in **huntingtin** (*HTT*) gene on chromosome **4** (**4 letters**) → toxic gain of function.

Symptoms typically manifest between age 30 and 50: chorea, athetosis, aggression, depression, dementia (sometimes initially mistaken for substance use).

Atrophy of caudate and putamen with ex vacuo ventriculomegaly.

↑ dopamine, ↓ GABA, ↓ ACh in brain.

Neuronal death via NMDA receptor binding and glutamate excitotoxicity.

Anticipation results from expansion of **CAG** repeats. **C**audate loses **A**Ch and **G**ABA.

Dementia

Decline in cognitive ability (eg, memory, executive function) with intact consciousness. Reversible causes of dementia include depression (pseudodementia), hypothyroidism, vitamin B₁₂ deficiency, neurosyphilis, normal pressure hydrocephalus.

Neurodegenerative

Alzheimer disease

Most common cause of dementia in older adults. Advanced age is the strongest risk factor. Down syndrome patients have ↑ risk of developing early-onset Alzheimer disease, as amyloid precursor protein (APP) is located on chromosome 21. ↓ ACh in brain.

Associated with the following altered proteins:

- ApoE-2: ↓ risk of sporadic form
- ApoE-4: ↑ risk of sporadic form
- APP, presenilin-1, presenilin-2: familial forms (10%) with earlier onset

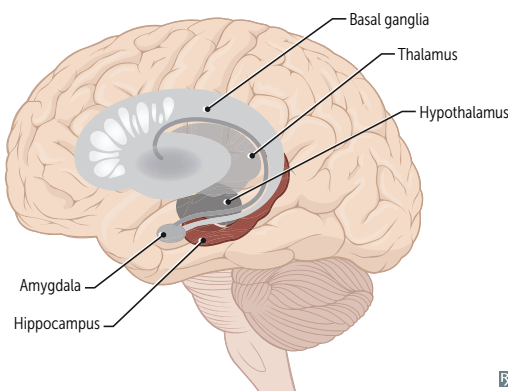
ApoE-2 is “**pro**two**ctive**,” ApoE-4 is “**four**” Alzheimer disease.

Widespread cortical atrophy, especially in hippocampi. Narrowing of gyri and widening of sulci.

Senile plaques **A** in gray matter: extracellular amyloid- β (A β) core; may cause amyloid angiopathy → intraparenchymal hemorrhage; A β is derived from cleavage of APP.

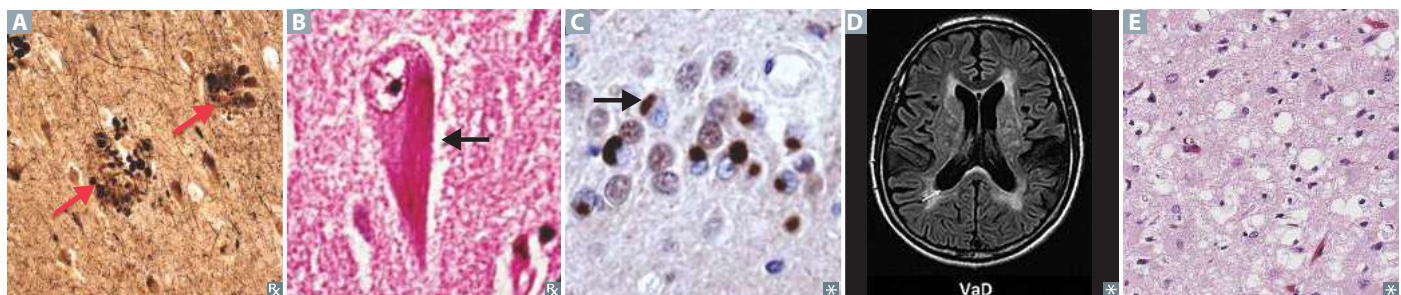
Neurofibrillary tangles **B**: intracellular, hyperphosphorylated tau protein = insoluble cytoskeletal elements; number of tangles correlates with degree of dementia.

Hirano bodies: intracellular eosinophilic proteinaceous rods in hippocampus.



Dementia (continued)

Frontotemporal dementia	Formerly called Pick disease. Early changes in personality and behavior (behavioral variant), or aphasia (primary progressive aphasia). May have associated movement disorders.	Frontal and/or temporal lobe atrophy. Inclusions of hyperphosphorylated tau (round Pick bodies C) or ubiquitinated TDP-43.
Lewy body dementia	Visual hallucinations (“ha Lewy cinations”), dementia with fluctuating cognition/alertness, REM sleep behavior disorder, and parkinsonism.	Intracellular Lewy bodies primarily in cortex. Called Lewy body dementia if cognitive and motor symptom onset < 1 year apart, otherwise considered dementia 2° to Parkinson disease.
Vascular		
Vascular dementia	2nd most common cause of dementia in older adults. Result of multiple arterial infarcts and/or chronic ischemia. Step-wise decline in cognitive ability with late-onset memory impairment.	MRI or CT shows multiple cortical and/or subcortical infarcts D .
Infective		
Creutzfeldt-Jakob disease	Rapidly progressive (weeks to months) dementia with myoclonus (“startle myoclonus”) and ataxia. Fatal. Caused by prions: PrP ^c → PrP ^{sc} (β-pleated sheet resistant to proteases). Typically sporadic, but may be transmitted by contaminated materials (eg, corneal transplant, neurosurgical equipment).	Spongiform cortex E (vacuolation without inflammation). Associated with periodic sharp wave complexes on EEG and ↑ 14-3-3 protein in CSF.
HIV-associated dementia	Subcortical dysfunction associated with advanced HIV infection. Characterized by cognitive deficits, gait disturbance, irritability, depressed mood.	Diffuse gray matter and subcortical atrophy. Microglial nodules with multinucleated giant cells.



Idiopathic intracranial hypertension

Also called pseudotumor cerebri. ↑ ICP with no obvious structural findings on imaging. Risk factors include **female** sex, **T**etracyclines, **O**besity, vitamin **A** excess, **D**anazol (**f**emale **T**OAD). Associated with dural venous sinus stenosis. Findings: headache (exacerbated when lying down), tinnitus, diplopia (usually from CN VI palsy), no change in mental status. Impaired optic nerve axoplasmic flow → papilledema. Visual field testing shows enlarged blind spot and peripheral constriction. LP reveals ↑ opening pressure and provides temporary headache relief.

Treatment: weight loss, acetazolamide, invasive procedures for refractory cases (eg, CSF shunt placement, optic nerve sheath fenestration surgery for visual loss).

Hydrocephalus

↑ CSF volume → ventricular dilation +/- ↑ ICP.

Communicating**Communicating hydrocephalus**

↓ CSF absorption by arachnoid granulations (eg, arachnoid scarring post-meningitis) → ↑ ICP, papilledema, herniation. All ventricles are dilated.

Normal pressure hydrocephalus

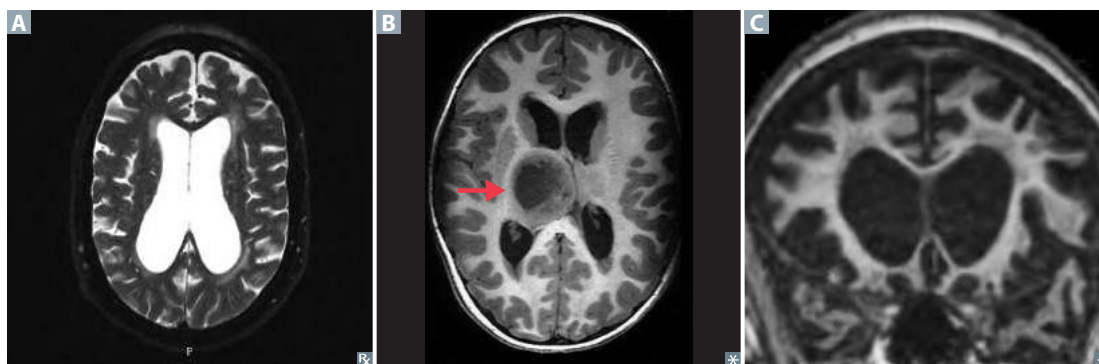
Affects older adults; idiopathic; CSF pressure elevated only episodically; does not result in increased subarachnoid space volume. Expansion of ventricles **A** distorts the fibers of the corona radiata → triad of **gait apraxia** (magnetic gait), **cognitive dysfunction**, and **urinary incontinence**. “**Wobbly, wacky, and wet.**” Treatment: CSF drainage via LP or shunt placement.

Noncommunicating (obstructive)**Noncommunicating hydrocephalus**

Caused by structural blockage of CSF circulation within ventricular system (eg, stenosis of aqueduct of Sylvius, colloid cyst blocking foramen of Monro, tumor **B**). Ventricles “upstream” of the obstruction are dilated.

Hydrocephalus mimics**Ex vacuo ventriculomegaly**

Appearance of ↑ CSF on imaging **C**, but is actually due to ↓ brain tissue and neuronal atrophy (eg, Alzheimer disease, HIV, frontotemporal dementia, Huntington disease). ICP is normal; NPH triad is not seen.



Multiple sclerosis

Autoimmune inflammation and demyelination of CNS (brain and spinal cord) with subsequent axonal damage. Most often affects females aged 20–40; higher prevalence in individuals who grew up farther from equator and have ↓ serum vitamin D levels. Can present with

- Optic neuritis (acute painful monocular visual loss, associated with relative afferent pupillary defect)
- Brainstem/cerebellar syndromes (eg, diplopia, ataxia, vertigo, scanning speech, dysarthria, intention tremor, nystagmus/intranuclear ophthalmoplegia [INO] [bilateral > unilateral])
- Pyramidal tract demyelination (eg, weakness, spasticity)
- Spinal cord syndromes (eg, electric shock–like sensation originating from cervical flexion, transmitted along the spinal cord [Lhermitte sign], neurogenic bladder, paraparesis, sensory manifestations affecting the trunk or one or more extremities)

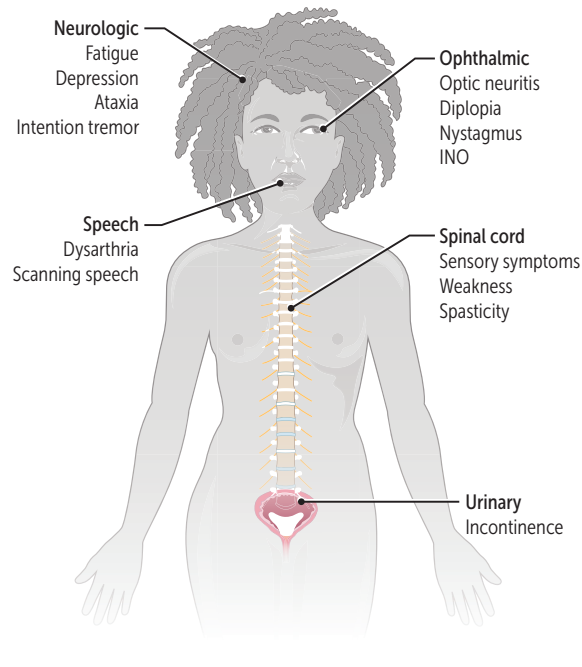
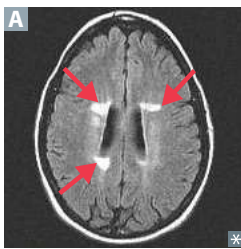
Symptoms may be exacerbated by stressors (eg, heat [Uhthoff phenomenon], exercise, or infection). Relapsing and remitting is most common clinical course.

FINDINGS

↑ IgG level and myelin basic protein in CSF. Oligoclonal bands in CSF aid in diagnosis. MRI is gold standard. Periventricular plaques **A** (areas of oligodendrocyte loss and reactive gliosis). Multiple white matter lesions disseminated in space and time.

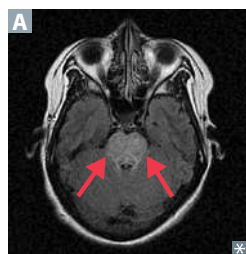
TREATMENT

Stop relapses and halt/slow progression with disease-modifying therapies (eg, β-interferon, glatiramer, natalizumab). Treat acute flares with IV steroids. Symptomatic treatment for neurogenic bladder (muscarinic antagonists, botulinum toxin injection), spasticity (baclofen, GABA_B receptor agonists), pain (TCAs, anticonvulsants).



Other demyelinating and dysmyelinating disorders

Osmotic demyelination syndrome



Also called central pontine myelinolysis. Massive axonal demyelination in pontine white matter

A 2° to rapid osmotic changes, most commonly iatrogenic correction of hyponatremia but also rapid shifts of other osmolytes (eg, glucose). Acute paralysis, dysarthria, dysphagia, diplopia, loss of consciousness. Can cause altered level of consciousness.

Correcting serum Na⁺ too fast:

- “From low to high, your pons will die” (osmotic demyelination syndrome)
- “From high to low, your brains will blow” (cerebral edema/herniation)

Acute inflammatory demyelinating polyneuropathy

Most common subtype of **Guillain-Barré syndrome**.

Autoimmune condition that destroys Schwann cells via inflammation and demyelination of motor fibers, sensory fibers, peripheral nerves (including CN III-XII). Likely facilitated by molecular mimicry and triggered by inoculations or stress. Despite association with infections (eg, *Campylobacter jejuni*, viruses [eg, Zika]), no definitive causal link to any pathogen.

Results in symmetric ascending muscle weakness/paralysis and depressed/absent DTRs beginning in lower extremities. Facial paralysis (usually bilateral) and respiratory failure are common. May see autonomic dysregulation (eg, cardiac irregularities, hypertension, hypotension) or sensory abnormalities. Most patients survive with good functional recovery.

↑ CSF protein with normal cell count (albuminocytologic dissociation).

Respiratory support is critical until recovery. Disease-modifying treatment: plasma exchange or IV immunoglobulins. No role for steroids.

Acute disseminated (postinfectious) encephalomyelitis

Multifocal inflammation and demyelination after infection or vaccination. Presents with rapidly progressive multifocal neurologic symptoms, altered mental status.

Charcot-Marie-Tooth disease

Also called hereditary motor and sensory neuropathy. Group of progressive hereditary nerve disorders related to the defective production of proteins involved in the structure and function of peripheral nerves or the myelin sheath. Typically autosomal dominant and associated with foot deformities (eg, pes cavus, hammer toe), lower extremity weakness (eg, foot drop), and sensory deficits (eg, decreased vibration, proprioception). Most common type, **CMT1A**, is caused by **PMP22** gene duplication (**Can't Move Toes**).

Progressive multifocal leukoencephalopathy



Demyelination of CNS **B** due to destruction of oligodendrocytes (2° to reactivation of latent JC virus infection). Associated with severe immunosuppression (eg, lymphomas and leukemias, AIDS, organ transplantation). Rapidly progressive, usually fatal. Predominantly involves parietal and occipital areas; visual symptoms are common. ↑ risk associated with organ transplantation, medications (eg, natalizumab).

Critical illness polyneuropathy

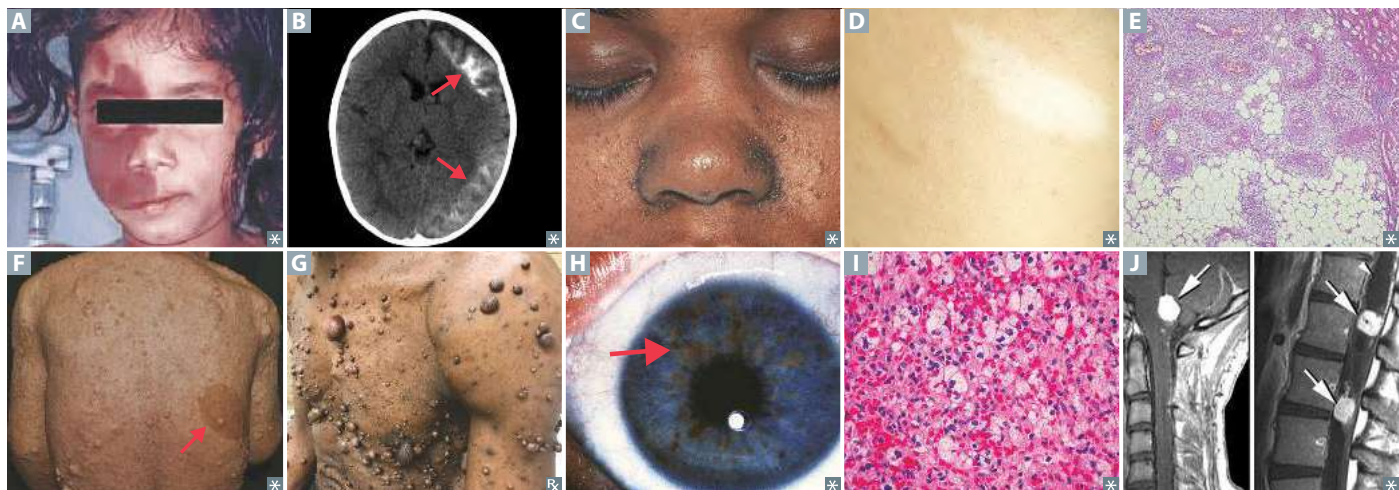
Axonal degeneration (likely from inflammatory mediators and microcirculation injury), ↓ nerve excitability 2° to Na⁺ channel inactivation → symmetric weakness (proximal > distal), ↓ deep tendon reflexes; diaphragmatic weakness may lead to difficulty weaning from mechanical ventilation.

Other disorders

Krabbe disease, metachromatic leukodystrophy, adrenoleukodystrophy.

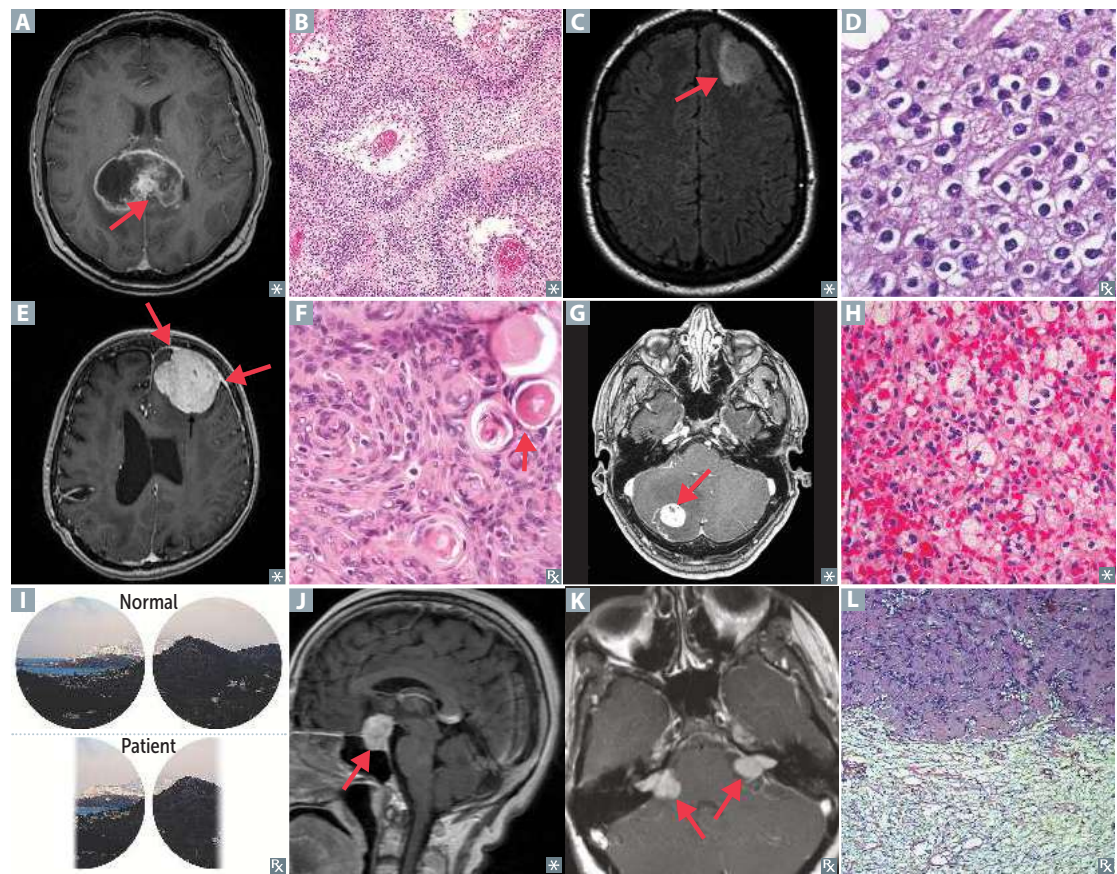
Neurocutaneous disorders

DISORDER	GENETICS	PRESENTATION	NOTES
Sturge-Weber syndrome	Congenital nonhereditary anomaly of neural crest derivatives. Somatic mosaicism of an activating mutation in one copy of the <i>GNAQ</i> gene.	Capillary vascular malformation → port-wine stain A (nevus flammeus or non-neoplastic birthmark) in CN V ₁ /V ₂ distribution; ipsilateral leptomeningeal angioma with calcifications B → seizures/epilepsy; intellectual disability; episcleral hemangioma → ↑ IOP → early-onset glaucoma.	Also called encephalotrigeminal angiomatosis.
Tuberous sclerosis complex	AD, variable expression. Mutation in tumor suppressor genes <i>TSC1</i> on chromosome 9 (hamartin), <i>TSC2</i> on chromosome 16 (tuberin; pronounce “twoberin”).	Hamartomas in CNS and skin, angiofibromas C , mitral regurgitation, ash-leaf spots D , cardiac rhabdomyoma, intellectual disability, renal angiomyolipoma E , seizures, shagreen patches.	↑ incidence of subependymal giant cell astrocytomas and ungual fibromas.
Neurofibromatosis type I	AD, 100% penetrance. Mutation in <i>NF1</i> tumor suppressor gene on chromosome 17 (encodes neurofibromin, a negative RAS regulator).	Café-au-lait spots F , Intellectual disability, Cutaneous neurofibromas G , Lisch nodules (pigmented iris hamartomas H), Optic gliomas, Pheochromocytomas, Seizures/focal neurologic Signs (often from meningioma), bone lesions (eg, sphenoid dysplasia).	Also called von Recklinghausen disease. 17 letters in “von Recklinghausen.” CICLOPSS .
Neurofibromatosis type II	AD. Mutation in <i>NF2</i> tumor suppressor gene (merlin) on chromosome 22.	Bilateral vestibular schwannomas, juvenile cataracts, meningiomas, ependymomas.	<i>NF2</i> affects 2 ears, 2 eyes.
von Hippel-Lindau disease	AD. Deletion of <i>VHL</i> gene on chromosome 3p. pVHL ubiquitinates hypoxia-inducible factor 1α.	Hemangioblastomas (high vascularity with hyperchromatic nuclei I) in retina, brainstem, cerebellum, spine J ; Angiomas; bilateral Renal cell carcinomas; Pheochromocytoma. HARP .	Numerous tumors, benign and malignant. VHL = 3 letters = chromosome 3; associated with RCC (also 3 letters).



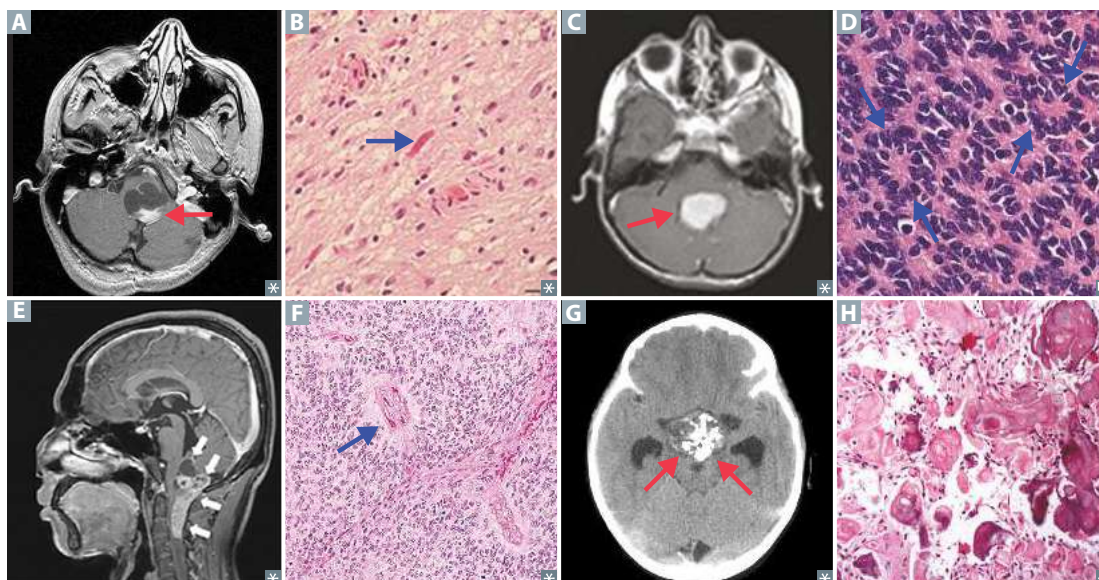
Adult primary brain tumors

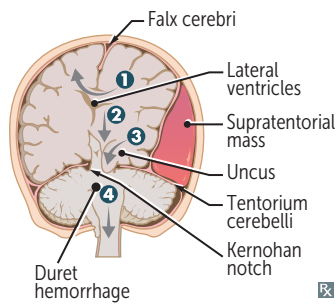
TUMOR	DESCRIPTION	HISTOLOGY
Glioblastoma	Common, highly malignant 1° brain tumor with ~ 1-year median survival. Found in cerebral hemispheres. Can cross corpus callosum (“butterfly glioma” A). Associated with <i>EGFR</i> amplification.	Astrocyte origin, GFAP ⊕. “Pseudopalisading” pleomorphic tumor cells B border central areas of necrosis, hemorrhage, and/or microvascular proliferation.
Oligodendroglioma	Relatively rare, slow growing. Most often in frontal lobes C . Often calcified.	Oligodendrocyte origin. “Fried egg” cells—round nuclei with clear cytoplasm D . “Chicken-wire” capillary pattern.
Meningioma	Common, typically benign. Females > males. Occurs along surface of brain or spinal cord. Extra-axial (external to brain parenchyma) and may have a dural attachment (“tail” E). Well circumscribed, spherical or lobular shape. Often asymptomatic; may present with seizures or focal neurologic signs. Treatment: resection and/or radiosurgery.	Arachnoid cell origin. Spindle cells concentrically arranged in a whorled pattern; psammoma bodies (laminated calcifications, arrow in F).
Hemangioblastoma	Most often cerebellar G . Associated with von Hippel-Lindau syndrome when found with retinal angiomas. Can produce erythropoietin → 2° polycythemia.	Blood vessel origin. Closely arranged, thin-walled capillaries with minimal intervening parenchyma H .
Pituitary adenoma	May be nonfunctioning (silent) or hyperfunctioning (hormone-producing). Nonfunctional tumors present with mass effect (eg, bitemporal hemianopia [due to pressure on optic chiasm I]). Pituitary apoplexy → hypopituitarism. Prolactinoma classically presents as galactorrhea, amenorrhea, ↓ bone density due to suppression of estrogen in females and as ↓ libido, infertility in males. Treatment: dopamine agonists (eg, bromocriptine, cabergoline), transsphenoidal resection.	Hyperplasia of only one type of endocrine cells found in pituitary. Most commonly from lactotrophs (prolactin) J → hyperprolactinemia. Less commonly, from somatotrophs (GH) → acromegaly, gigantism; corticotrophs (ACTH) → Cushing disease. Rarely, from thyrotrophs (TSH), gonadotrophs (FSH, LH).
Schwannoma	Classically at the cerebellopontine angle K , benign, involving CNs V, VII, and VIII, but can be along any peripheral nerve. Often localized to CN VIII in internal acoustic meatus → vestibular schwannoma (can present as hearing loss, tinnitus, and unsteady gait). Bilateral vestibular schwannomas found in NF-2. Treatment: resection or stereotactic radiosurgery.	Schwann cell origin, S-100 ⊕. Biphasic, dense, hypercellular areas containing spindle cells alternating with hypocellular, myxoid areas L .

Adult primary brain tumors (continued)

Childhood primary brain tumors

TUMOR	DESCRIPTION	HISTOLOGY
Pilocytic astrocytoma	Most common 1° brain tumor in childhood. Usually well circumscribed. In children, most often found in posterior fossa (eg, cerebellum). May be supratentorial. Cystic appearance with mural nodule A . Benign; good prognosis.	Astrocyte origin, GFAP ⊕. Bipolar neoplastic cells with hairlike projections. Associated with microcysts and Rosenthal fibers (eosinophilic, corkscrew fibers B).
Medulloblastoma	Most common malignant brain tumor in childhood. Commonly involves cerebellum C . Can involve the cerebellar vermis → truncal ataxia. Can compress 4th ventricle → noncommunicating hydrocephalus → headaches, papilledema. Can send “drop metastases” to spinal cord.	Form of primitive neuroectodermal tumor (PNET). Homer-Wright rosettes (small, round, blue cells surrounding central area of neuropil D). Synaptophysin ⊕.
Ependymoma	Most commonly found in 4th ventricle E → noncommunicating hydrocephalus. Poor prognosis.	Ependymal cell origin. Characteristic perivascular pseudorosettes F . Rod-shaped blepharoplasts (basal ciliary bodies) found near the nucleus.
Craniopharyngioma	Most common childhood supratentorial tumor. Calcification is common G . Commonly arises along pituitary stalk → compression of optic chiasm → bitemporal hemianopia (may be confused with pituitary adenoma). Associated with a high recurrence rate.	Derived from remnants of Rathke pouch (ectoderm). Anucleate squamous cells (“ghost cells”) forming keratinous nodules with dystrophic calcifications H . Cholesterol crystals found in “motor oil”-like fluid within tumor.
Pineal gland tumors	Most commonly extragonadal germ cell tumors. ↑ incidence in males. Present with noncommunicating hydrocephalus (compression of cerebral aqueduct), Parinaud syndrome (compression of dorsal midbrain)—triad of upward gaze palsy, convergence-retraction nystagmus, and light-near dissociation.	Similar to testicular seminomas.



Herniation syndromes

1 Cingulate (subfalcine) herniation under falx cerebri

Can cause anterior cerebral artery compression → contralateral lower extremity weakness. If affecting the dominant hemisphere, can cause arcuate fasciculus compression → aphasia.

2 Central/downward transtentorial herniation

Caudal displacement of brainstem → rupture of paramedian basilar artery branches → Duret hemorrhages. Usually fatal.

3 Uncal transtentorial herniation

Uncus = medial temporal lobe. Early herniation → ipsilateral blown pupil (unilateral CN III compression), contralateral hemiparesis. Late herniation → coma, Kernohan phenomenon (misleading ipsilateral hemiparesis +/- contralateral blown pupil due to contralateral cerebral peduncle +/- CN III compression against Kernohan notch).

4 Cerebellar tonsillar herniation into the foramen magnum

Coma and death result when these herniations compress the brainstem.

Motor neuron signs

SIGN	UMN LESION	LMN LESION	COMMENTS
Weakness	+	+	Lower motor neuron = everything lowered (↓ muscle mass, tone, reflexes, toes)
Atrophy	–	+	
Fasciculations	–	+	
Reflexes	↑	↓	Upper motor neuron = everything up (↑ tone, reflexes, toes)
Tone	↑	↓	
Babinski	+	–	
Spastic paresis	+	–	Fasciculations = muscle twitching Positive Babinski is normal in infants
Flaccid paralysis	–	+	
Clasp knife spasticity	+	–	

Spinal cord lesions

Poliomyelitis



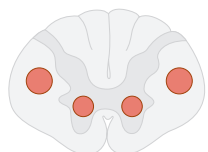
Destruction of anterior horns by poliovirus. Fecal-oral transmission → replication in lymphoid tissue of oropharynx and small intestine → spread to CNS via bloodstream. Acute LMN signs (**asymmetric** weakness) and symptoms of viral meningitis (eg, fever, headache, neck stiffness). Respiratory muscle involvement leads to respiratory failure. CSF shows ↑ WBCs (lymphocytic pleocytosis) and slight ↑ of protein (with no change in CSF glucose). Poliovirus can be isolated from stool or throat secretions.

Spinal muscular atrophy



Congenital degeneration of anterior horns. Autosomal recessive **SMN1** mutation (encodes **survival motor neuron protein**) → defective snRNP assembly → LMN apoptosis. Spinal muscular atrophy type 1 (most common) is also called **Werdnig-Hoffmann disease**. LMN signs only (**symmetric** weakness). “Floppy baby” with marked hypotonia (flaccid paralysis) and tongue fasciculations.

Amyotrophic lateral sclerosis



Combined UMN (corticospinal/corticobulbar) and LMN (brainstem/spinal cord) degeneration. Usually idiopathic. Familial form (less common) may be linked to **SOD1** mutations (encodes **superoxide dismutase 1**). ALS is also called **Lou Gehrig disease**. LMN signs: flaccid limb weakness, fasciculations, atrophy, bulbar palsy (dysarthria, dysphagia, tongue atrophy). UMN signs: spastic limb weakness, hyperreflexia, clonus, pseudobulbar palsy (dysarthria, dysphagia, emotional lability). No sensory or bowel/bladder deficits. Fatal (most often from respiratory failure). Treatment: riluzole (“**riLouzole**”), edaravone (free radical scavenger) → slow functional decline.

Tabes dorsalis



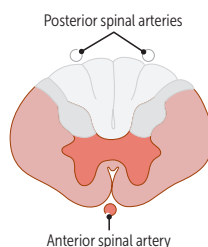
Degeneration/demyelination of dorsal columns and roots by *T pallidum* (3° syphilis). Causes progressive sensory ataxia (impaired proprioception → poor coordination). ⊕ Romberg sign and absent DTRs. Associated with shooting pain, Argyll Robertson pupils, Charcot joints.

Subacute combined degeneration

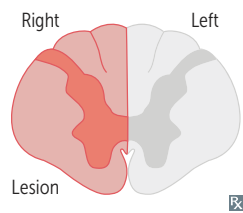


Demyelination of **S**pinocerebellar tracts, lateral **C**orticospinal tracts, and **D**orsal columns (**SCD**) due to vitamin B₁₂ deficiency. Ataxic gait, paresthesias, impaired position/vibration sense (⊕ Romberg sign), UMN signs.

Anterior spinal artery occlusion

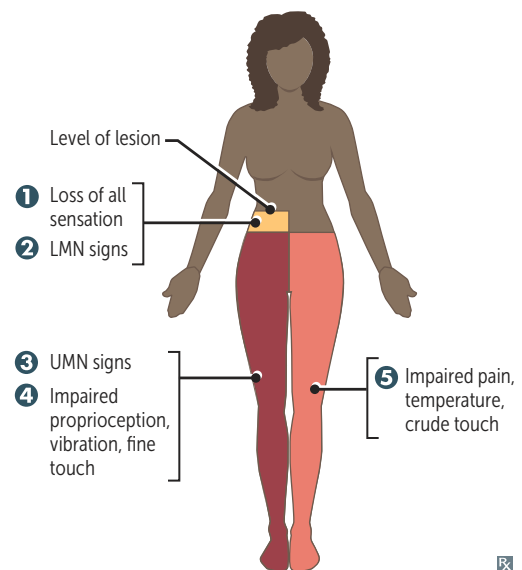


Spinal cord infarction sparing dorsal horns and dorsal columns. Watershed area is mid-thoracic ASA territory, as the artery of Adamkiewicz supplies ASA below T8. Can be caused by aortic aneurysm repair. Presents with UMN signs below the lesion (corticospinal tract), LMN signs at the level of the lesion (anterior horn), and loss of pain and temperature sensation below the lesion (spinothalamic tract).

Brown-Séquard syndrome

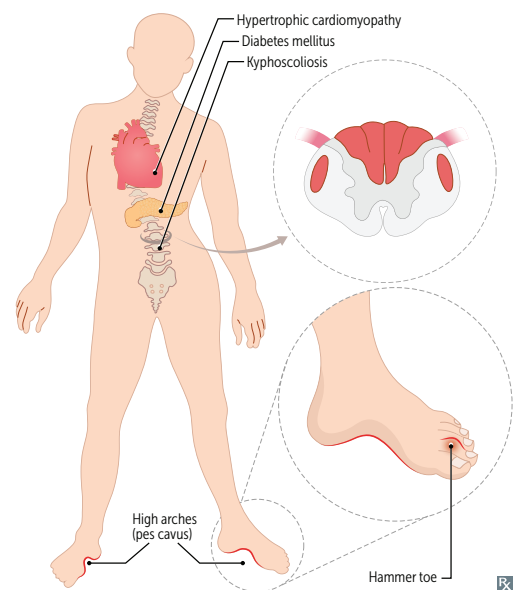
Hemisection of spinal cord. Findings due to deficits in

- ➊ All sensory pathways—ipsilateral loss of all sensation **at** the lesion level.
 - ➋ Corticospinal tract—ipsilateral LMN signs (eg, flaccid paralysis) **at** lesion level.
 - ➌ Corticospinal tract—ipsilateral UMN signs **below** lesion level.
 - ➍ Dorsal columns—ipsilateral loss of proprioception, vibration, and fine (2-point discrimination) touch **below** lesion level.
 - ➎ Spinothalamic tract—contralateral loss of pain, temperature, and crude (nondiscriminative) touch **below** lesion level.
- Oculosympathetic pathway (if lesion occurs above T1)—ipsilateral Horner syndrome.

**Friedreich ataxia**

Autosomal recessive trinucleotide repeat disorder (**GAA**)_n on chromosome 9 in gene that encodes frataxin (iron-binding protein). Leads to impairment in mitochondrial functioning. Degeneration of lateral corticospinal tract (spastic paralysis), spinocerebellar tract (ataxia), dorsal columns (↓ vibratory sense, proprioception), and dorsal root ganglia (loss of DTRs). **Staggering** gait, frequent **falling**, nystagmus, dysarthria, pes cavus, hammer toes, **diabetes** mellitus, **hypertrophic cardiomyopathy** (cause of death). Presents in childhood with kyphoscoliosis **A**.

Friedreich is **frataxic** (**frataxin**): he's your favorite **frat** brother, always **staggering** and **falling** but has a **sweet, big heart**. Ataxic **GAA**it.

**Cerebral palsy**

Permanent motor dysfunction resulting from nonprogressive injury to developing fetal/infant brain. Most common movement disorder in children.

Multifactorial etiology; prematurity and low birth weight are the strongest risk factors. Associated with development of periventricular leukomalacia (focal necrosis of white matter tracts).

Presents with UMN signs (eg, spasticity, hyperreflexia) affecting ≥ 1 limbs, persistence of primitive reflexes, abnormal posture, developmental delay in motor skills, neurobehavioral abnormalities (excessive docility, irritability).

Treatment: muscle relaxants (eg, baclofen), botulinum toxin injections, selective dorsal rhizotomy. Prevention: prenatal magnesium sulfate for high-risk pregnancies ↓ incidence and severity.

Common cranial nerve lesions

CN V motor lesion	Jaw deviates toward side of lesion due to unopposed force from the opposite pterygoid muscle.
CN X lesion	Uvula deviates away from side of lesion. Weak side collapses and uvula points away.
CN XI lesion	Weakness turning head away from side of lesion (SCM). Shoulder droop on side of lesion (trapezius).
CN XII lesion	LMN lesion. Tongue deviates toward side of lesion (“lick your wounds”) due to weakened tongue muscles on affected side.

Facial nerve lesions



Bell palsy is the most common cause of peripheral facial palsy **A**. Usually develops after HSV reactivation. Treatment: glucocorticoids +/- acyclovir. Most patients gradually recover function, but aberrant regeneration can occur. Other causes of peripheral facial palsy include Lyme disease, herpes zoster (**Ramsay Hunt syndrome**—triad of ipsilateral facial paralysis, otalgia, and vesicles near the auditory canal), sarcoidosis, tumors (eg, parotid gland), diabetes mellitus.

	Upper motor neuron lesion	Lower motor neuron lesion
LESION LOCATION	Motor cortex, connection from motor cortex to facial nucleus in pons	Facial nucleus, anywhere along CN VII
AFFECTED SIDE	Contralateral	Ipsilateral
MUSCLES INVOLVED	Lower muscles of facial expression	Upper and lower muscles of facial expression
FOREHEAD INVOLVEMENT	Spared, due to bilateral UMN innervation	Affected
OTHER SYMPTOMS	Variable; depends on size of lesion	Incomplete eye closure (dry eyes, corneal ulceration), hyperacusis, loss of taste sensation to anterior tongue

